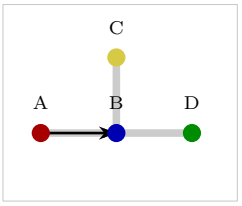
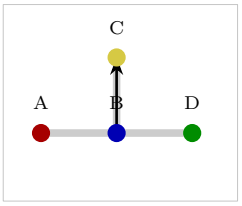
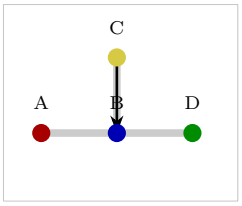
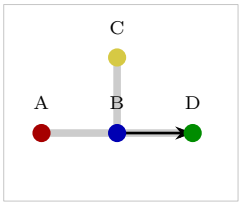
Move a pawn on the roads. Keep each journey as ordered step tiles. Two adjacent steps along the same road in opposite directions are an out-and-back pair. Remove only out-and-back pairs, and replay the other steps in order. The map, start, and finish stay fixed.

input



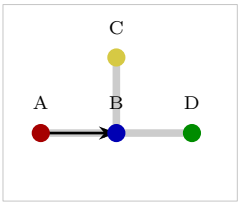


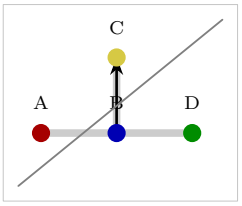


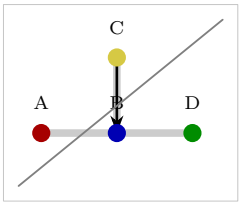


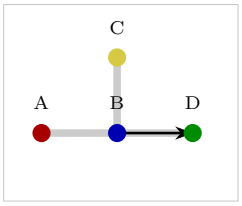
$A \to B \to C \to B \to D$

remove

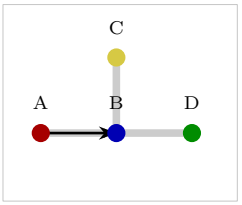


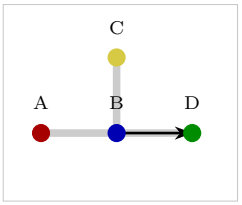






output





$A \to B \to D$

No steps left means the pawn stays at its start.

Problem 1: Shorten each recorded trip until no move is possible. Make a different seven-step trip from C to F that finishes with the same route as the last one.

$A \to B \to A \to B \to D \to F \to D \to E$

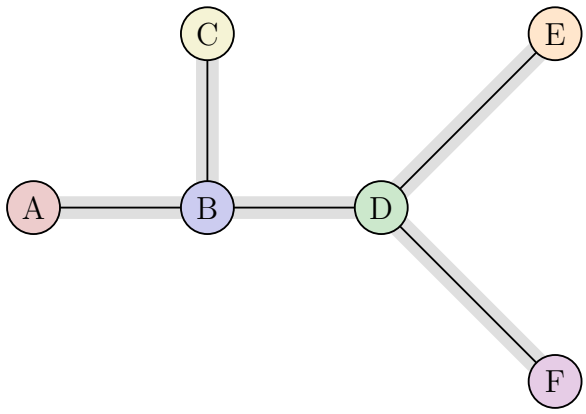
$\longrightarrow$ _____

$A \to B \to D \to F \to D \to E$

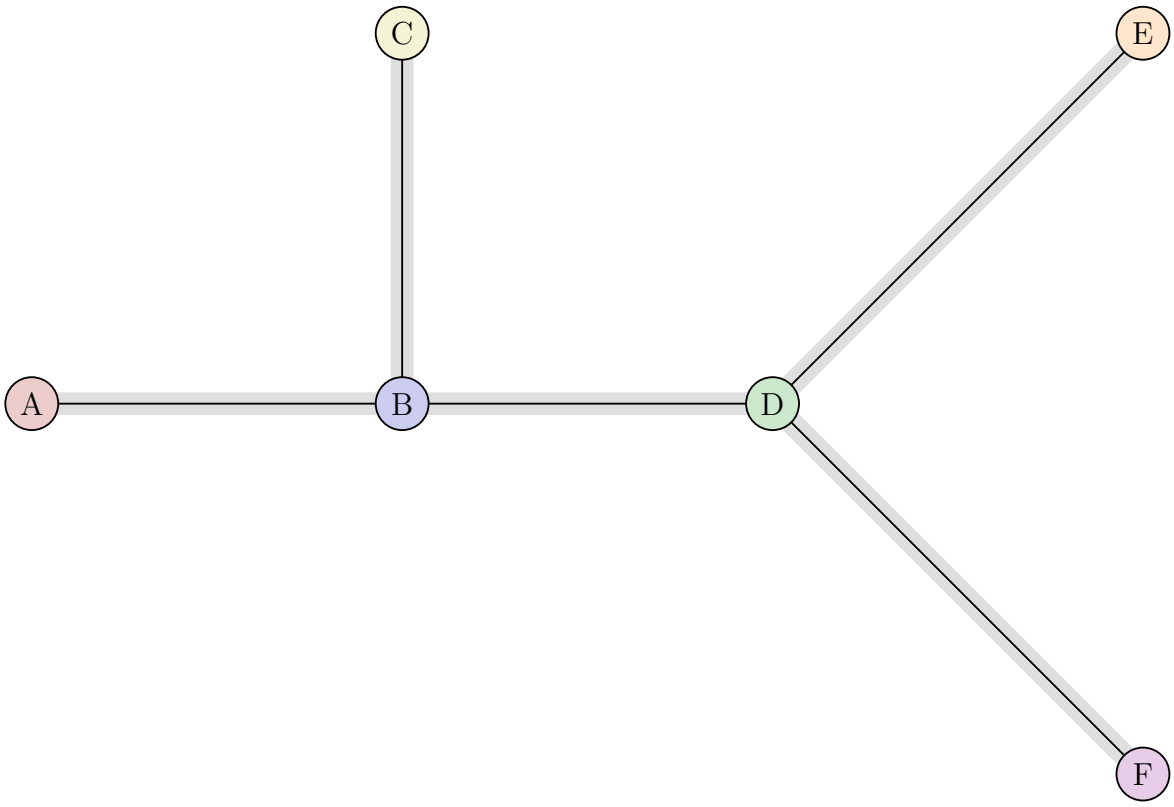
$\longrightarrow$ _____

$C \to B \to C \to B \to D \to F$

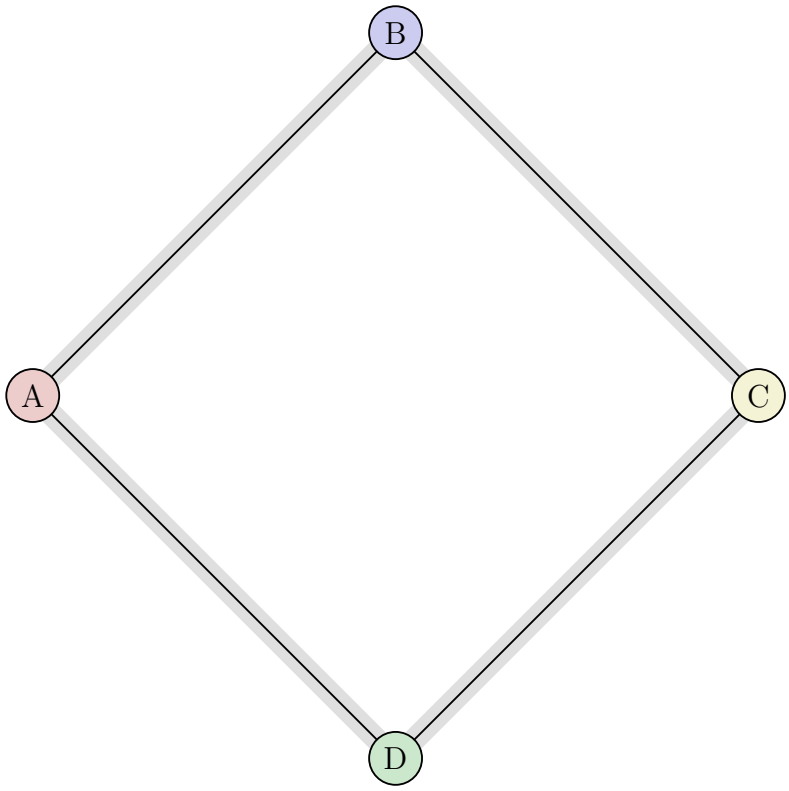
$\longrightarrow$ _____



Problem 2: Make two different 12-step trips from A back to A, each using every road. Can any trip home on this map survive after every out-and-back pair is removed?



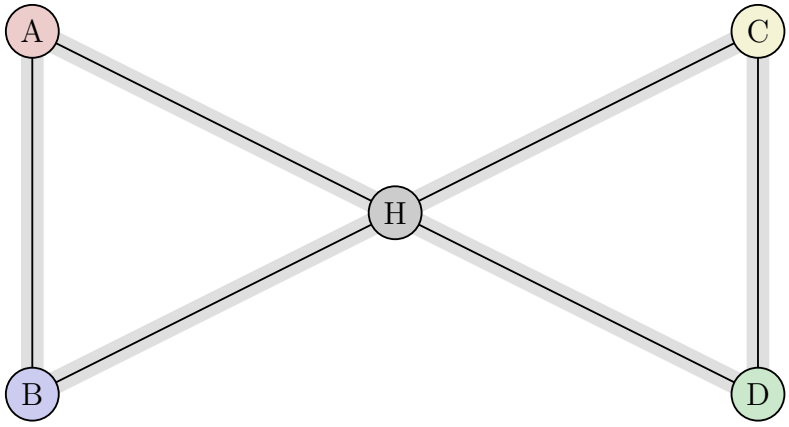
Problem 3: Make every different trip from A back to A that can remain after shortening an eight-step ring trip. Find the fewest road steps in a surviving trip home.



Problem 4: Remove different first out-and-back pairs from the recorded trip below. Can two removal orders end differently?

$$A \to B \to A \to B \to C \to B \to A \to D \to C \to D \to A \quad \longrightarrow _____$$

Problem 5: Left forward (LF) is H to A to B and back to H, left backward (LB) is H to B to A and back to H, and right forward (RF) is H to C to D and back to H. Join these three ring trips in each of the six possible orders and shorten each journey. Which orders shorten to the same trip?



Problem 6: Make two trips from H back to H, each using both rings and at least 12 road steps. One must shorten to staying at H; the other must lose no steps.